End-Point Security

Sub-Topics:

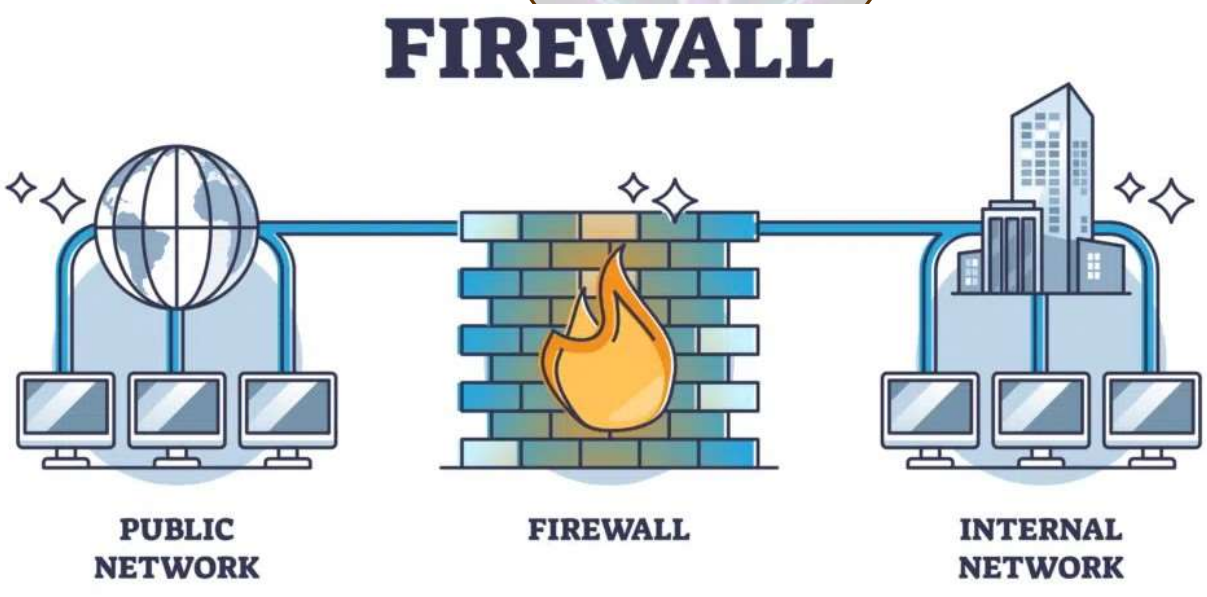
Firewall Architecture & Types, Firewall Rules & Policies, DMZ, (iptables, UFW, pfSense), Bypassing, Intrusion Detection & Prevention Systems (IDS/IPS) with Detection Techniques (Signature, Anomaly, Behavioral), and Countermeasures.

What is a Firewall?

A **Firewall** is a **network security system** that monitors and controls incoming and outgoing network traffic based on predefined security rules.

It acts as a **barrier between a trusted internal network and an untrusted external network (such as the Internet)**.

The main purpose of a firewall is to **prevent unauthorized access, cyber attacks, malware, and data theft** while allowing legitimate communication.



Packet Filtering Firewall Architecture

Is architecture me firewall **network packets ko inspect karta hai** aur predefined rules ke basis par allow ya block karta hai.

Check kiye jane wale parameters:

- Source IP Address
- Destination IP Address
- Port Number
- Protocol (TCP, UDP, ICMP)



Advantages

- Fast performance
- Simple implementation

Disadvantages

- Deep inspection nahi karta
- Advanced attacks detect nahi kar pata

Dual-Homed Host Architecture

Is architecture me ek **single host firewall** hota hai jisme **2 network interfaces (NIC)** hote hain.

- Ek interface **internal network** se connected hota hai
- Dusra **external network (internet)** se connected hota hai

Traffic ko **direct route nahi milta**, sab traffic firewall host ke through hi pass hota hai.

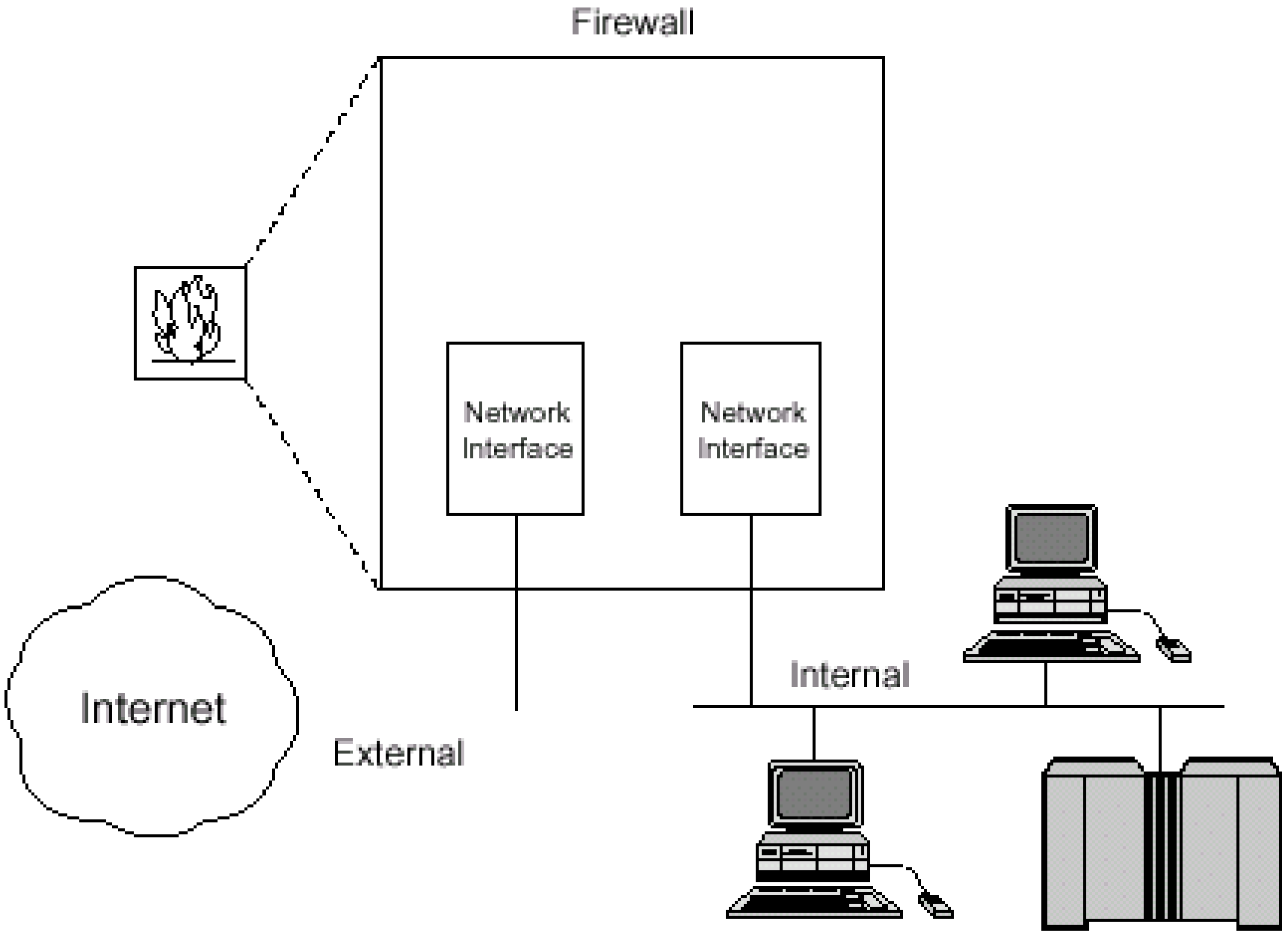


Advantages

- More secure than packet filtering
- Internal network ko direct access nahi milta

Disadvantages

- Single point of failure
- High load ho sakta hai


Firewall
Network Interface
Network Interface
Internet
External
Internal

Screened Host Architecture

Is architecture me **router firewall + bastion host** use hota hai.
Components:

- **Packet filtering router**
- **Bastion Host** (secure server)

Process:
1.Router suspicious traffic ko filter karta hai
2.Bastion host application level security provide karta hai

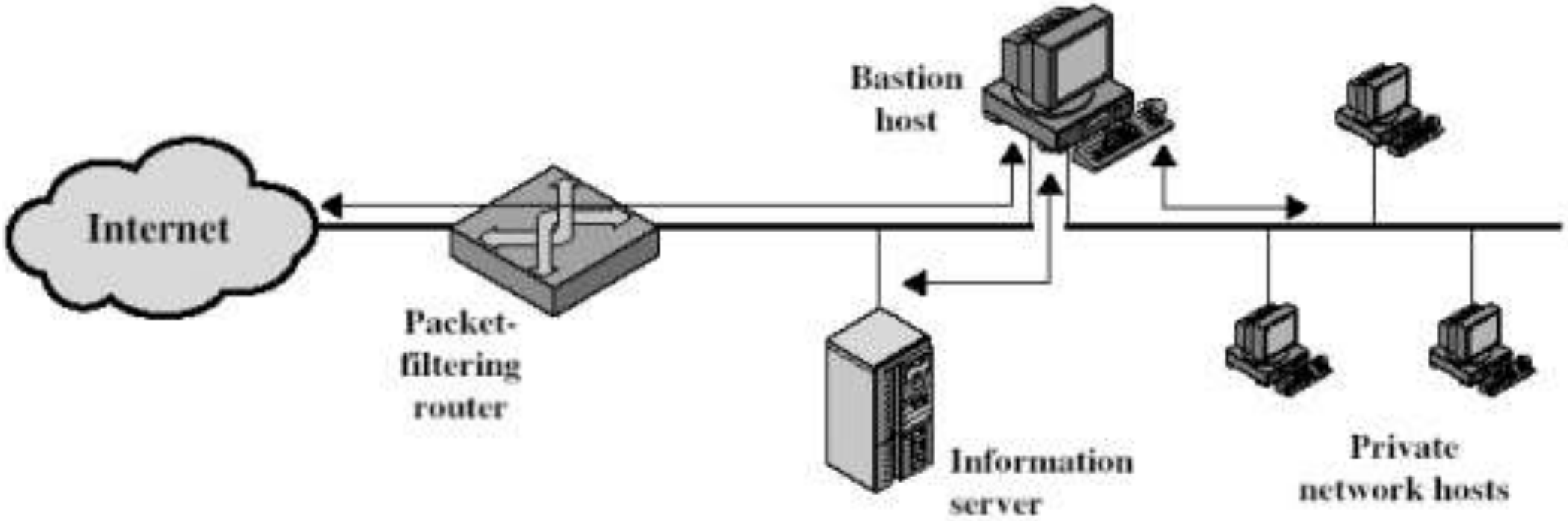


Advantages

- Dual security layer
- More secure than basic firewall

Disadvantages

- Configuration complex hota hai



(b) Screened host firewall system (dual-homed bastion host)

Screened Subnet Architecture (DMZ Architecture)

Ye **sabse secure firewall architecture** mana jata hai.
Isme **DMZ (Demilitarized Zone)** create kiya jata hai jahan public servers rakhe jate hain.

Example servers:

- Web Server
- Mail Server
- DNS Server



Advantages

- High security
- Internal network directly exposed nahi hota

Disadvantages

- Costly
- Complex configuration

Types of Firewall

Firewalls can be classified based on **how they inspect and filter network traffic**.

- Packet Filtering Firewall
- Stateful Inspection Firewall
- Proxy Firewall (Application-Level Gateway)
- Next Generation Firewall (NGFW)



Stateful Inspection Firewall

A **Stateful Inspection Firewall** monitors the **state of active connections** and keeps track of sessions in a **state table**.
Instead of checking each packet independently, it verifies whether the packet belongs to an **existing trusted session**.

Example:
When a user opens a website:

- The firewall records the connection.
- It allows response packets from the web server because they belong to the same session.



Advantages

- More secure than packet filtering
- Tracks connection states
- Better control over traffic

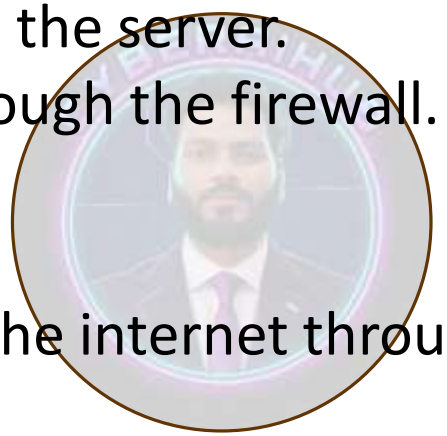
Disadvantages

- Uses more memory and CPU
- More complex configuration

Proxy Firewall (Application-Level Gateway)

A **Proxy Firewall** works at the **Application Layer (Layer 7)** of the OSI model.
It acts as an **intermediary between users and the internet**.
Instead of direct communication:

- The client sends a request to the proxy firewall.
- The firewall forwards the request to the server.
- The server response is returned through the firewall.



Example:
In many universities, students access the internet through a **proxy server** that filters websites.

Advantages

- Very high security
- Can inspect application data
- Hides internal network IP addresses

Disadvantages

- Slower performance
- Higher resource consumption

Next Generation Firewall (NGFW)

A **Next Generation Firewall (NGFW)** is an advanced firewall that combines **traditional firewall features with modern security technologies**.

It includes:

- Deep Packet Inspection (DPI)
- Intrusion Detection/Prevention (IDS/IPS)
- Application awareness
- Malware protection
- SSL inspection

Example:

Companies use NGFW products like:

- Cisco Firepower
- Palo Alto Firewall
- Fortinet FortiGate



Advantages

- Very strong security
- Detects modern cyber attacks
- Supports advanced threat intelligence

Disadvantages

- Expensive
- Complex configuration

These firewalls can **detect hacking attempts and malware in real time**.

Firewall Rules and Policies

Firewall Rules wo specific instructions hoti hain jo firewall ko batati hain ki **kaunsa network traffic allow karna hai aur kaunsa block karna hai**. Ye rules firewall ke andar configure kiye jate hain aur **packets ko filter karne ke liye use hote hain**.



Firewall rules generally **packet information** ke basis par kaam karte hain, jaise:

- Source IP Address
- Destination IP Address
- Source Port Number
- Destination Port Number
- Protocol (TCP, UDP, ICMP)
- Direction (Inbound / Outbound)

Firewall jab koi packet receive karta hai to wo **rules list ko top se bottom tak check karta hai**. Agar packet kisi rule se match ho jata hai to us rule ke according **Allow, Deny, Reject ya Drop action perform hota hai**.

Example of Firewall Rule

Rule No	Source IP	Destination IP	Port	Protocol	Action
1	Any	Web Server	80	TCP	Allow
2	Any	Any	23	TCP	Block
3	Internal Network	Internet	443	TCP	Allow



Explanation:

- Rule 1: Users web server ko HTTP se access kar sakte hain.
- Rule 2: Telnet traffic completely blocked hai.
- Rule 3: Internal users HTTPS websites open kar sakte hain.

Firewall Policy ek **set of firewall rules ka collection** hota hai jo network security ko manage karta hai. Ye decide karta hai ki **overall network traffic ko kaise control karna hai**.
Simple words me:
Policy = Multiple Firewall Rules + Security Strategy

Firewall policy define karti hai:

- Kaun network resources access kar sakta hai
- Kaunse ports open honge
- Kaunse protocols allowed honge
- Kaunse traffic ko block karna hai
- Internal aur external communication kaise control hoga



DMZ (Demilitarized Zone)

DMZ (Demilitarized Zone) ek **special network segment** hota hai jo **internal network (LAN) aur external network (Internet)** ke beech me placed hota hai. Iska purpose hota hai **public services ko expose karna without directly exposing internal network**.

Simple words me:

☞ DMZ ek **buffer zone** hai jo attackers ko **internal network tak direct access se rokta hai**.



DMZ usually do tarah se implement hota hai:

1. Two-Firewall DMZ

- Firewall 1 → Internet aur DMZ ke beech
- Firewall 2 → DMZ aur Internal Network ke beech

2. Three-Legged Firewall (Single Firewall DMZ)

- Ek hi firewall ke 3 interfaces hote hain:
  - External (Internet)
  - DMZ
  - Internal Network

DMZ me kaunse servers hote hain?

DMZ me wo servers rakhe jate hain jo **publicly accessible hone chahiye**:

- Web Server (Website hosting)
- Mail Server (Email services)
- DNS Server
- FTP Server

⚠ Important:

Sensitive systems jaise **database server, internal files, HR system** DMZ me nahi rakhe jate.



👉 Agar DMZ server hack bhi ho jaye, attacker **internal network tak directly nahi pahunch sakta**.

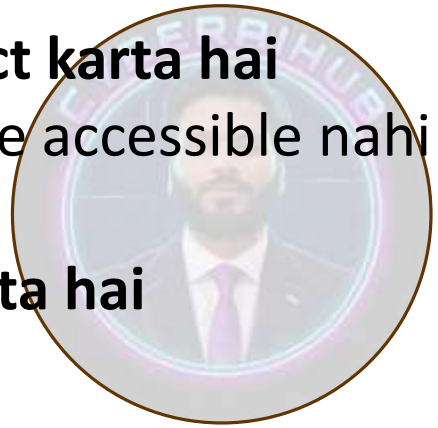
Example (Real Life)

Ek **Banking System** ka example:

- DMZ me: Web server (online banking website)
- Internal network me: Database server (customer data)

Customer jab website open karta hai:

- Wo sirf **DMZ web server se interact karta hai**
- Database server directly internet se accessible nahi hota



☞ Isse **data breach ka risk kam ho jata hai**

Advantages of DMZ

- High level security
- Internal network hidden rehta hai
- Public services safely provide kar sakte hain
- Attack impact limited ho jata hai

Disadvantages of DMZ

- Setup complex hota hai
- Cost zyada hoti hai
- Proper configuration zaroori hai

Iptables:



iptables Linux ka **command-line firewall tool** hai jo kernel ke **Netfilter framework** par based hota hai. Ye system level par **packet filtering aur NAT (Network Address Translation)** handle karta hai.

Advantages

- Powerful and flexible
- Deep control over traffic
- Widely used in Linux servers

Disadvantages

- Complex syntax
- Beginners ke liye difficult

Key Concepts

- **Tables**: filter, nat, mangle
- **Chains**: INPUT, OUTPUT, FORWARD
- **Rules**: allow/deny conditions

Working

1. Packet system me enter hota hai
2. Appropriate chain (INPUT/OUTPUT/FORWARD) select hoti hai
3. Rules top-to-bottom check hote hain
4. Match hone par action (ACCEPT, DROP, REJECT) apply hota hai




```
# Allow SSH
iptables -A INPUT -p tcp --dport 22 -j ACCEPT
# Block Telnet
iptables -A INPUT -p tcp --dport 23 -j DROP
# Save rules
iptables-save
```

UFW (Uncomplicated Firewall)

UFW (Uncomplicated Firewall) ek **user-friendly frontend** hai jo internally **iptables** ko use karta hai. Ye mainly **Ubuntu systems** me use hota hai.

Purpose

👉 iptables ko simplify karna



```
# Enable firewall
• ufw enable
# Allow SSH
• ufw allow 22
# Allow HTTP
• ufw allow 80
# Deny Telnet
• ufw deny 23
# Check status
• ufw status
```

Agar aapko sirf web server run karna hai:
•Allow: 80 (HTTP), 443 (HTTPS)
•Block: baaki sab

Advantages
•Easy to use
•Simple commands
•Quick configuration



Disadvantages
•Less flexibility compared to iptables
•Advanced control limited

pfSense Firewall

pfSense ek **open-source firewall and router software** hai jo **FreeBSD** par based hai. Ye ek **full-featured network security solution** provide karta hai.

Features

- Web-based GUI (no command line needed)
- Firewall + Router + VPN
- Intrusion Detection/Prevention (Snort/Suricata)
- Traffic monitoring
- VLAN support



Working

- pfSense ek dedicated machine ya VM par install hota hai
- Ye **network gateway firewall** ki tarah kaam karta hai
- Saare incoming/outgoing traffic ko filter karta hai

Example

Ek office setup:

•pfSense router ke jagah use hota hai

•Rules:

- Allow web traffic (80/443)
- Block suspicious IPs
- VPN access for employees



Advantages

•Very powerful

•GUI based (easy management)

•Enterprise-level features free me

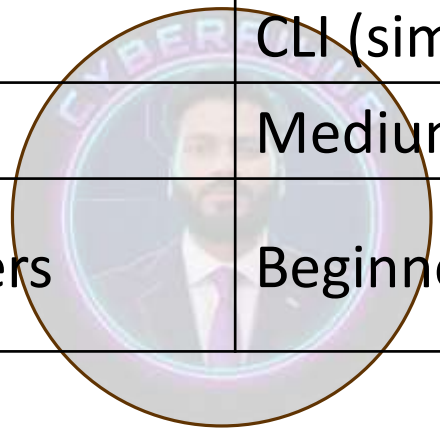
Disadvantages

•Setup thoda complex ho sakta hai

•Hardware required

Final Comparison

Feature	iptables	UFW	pfSense
Type	Command-line tool	Frontend for iptables	Full firewall system
Ease of Use	Hard	Easy	Medium
Interface	CLI	CLI (simple)	Web GUI
Flexibility	High	Medium	Very High
Use Case	Advanced users	Beginners	Enterprise / Network level



IDS (Intrusion Detection System)

IDS (Intrusion Detection System) ek **security monitoring system** hai jo network ya system me **suspicious activities, attacks, ya policy violations detect karta hai**.

☞ IDS **alert generate karta hai**, lekin traffic ko block nahi karta.

Types of IDS

1.NIDS (Network IDS)

1. Network traffic ko monitor karta hai
2. Example: Snort



2.HIDS (Host IDS)

1. Individual system (host) par monitor karta hai
2. Example: OSSEC

Working

1.Network/system activity monitor hoti hai

2.Suspicious pattern detect hota hai

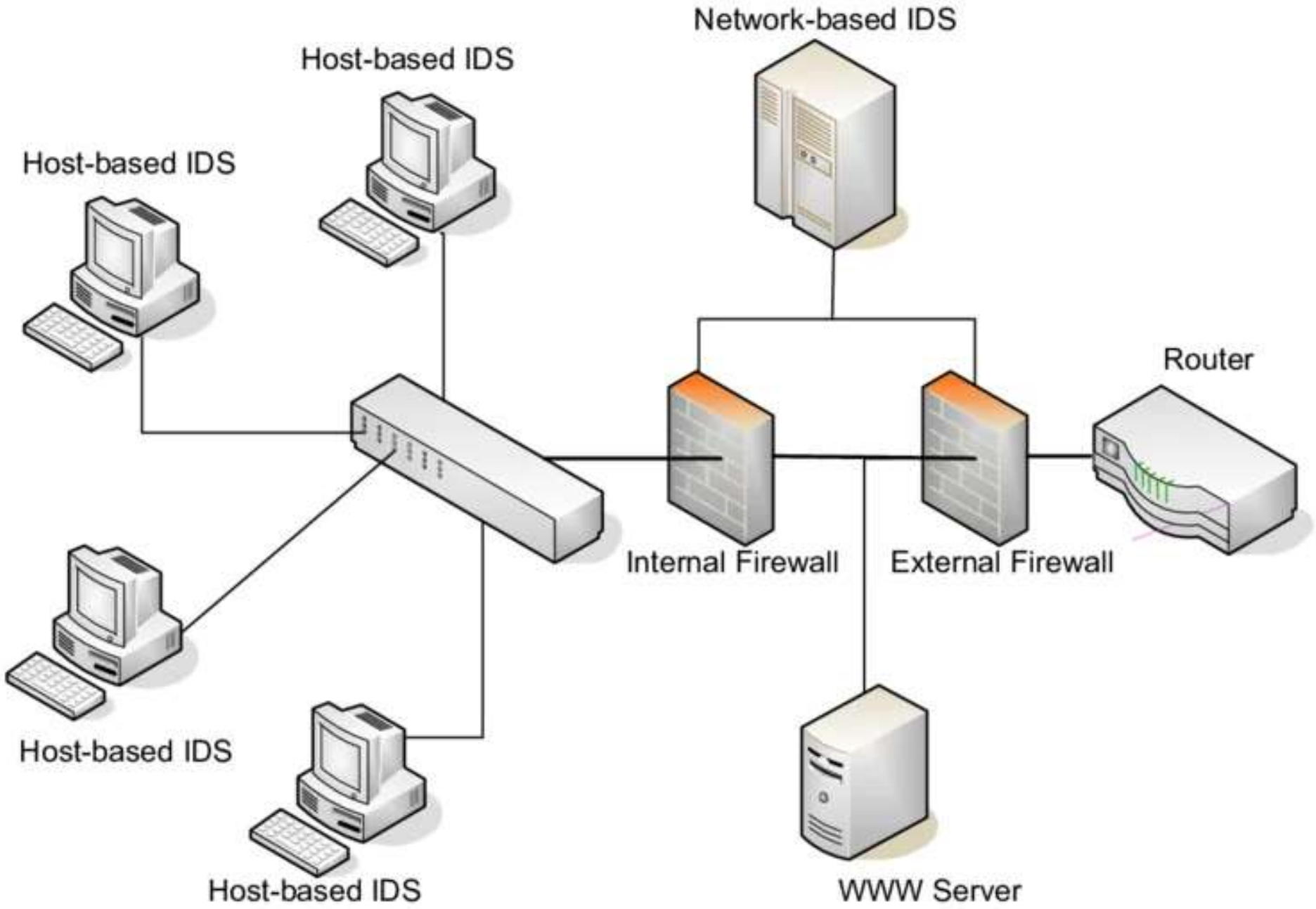
3.Alert generate hota hai (log, email, SIEM)

Example

Agar koi attacker multiple login attempts karta hai (brute force), IDS:

•Attack detect karega

•Admin ko alert bhejega


Network-based IDS
Host-based IDS
Host-based IDS
Router
Internal Firewall
External Firewall
Host-based IDS
Host-based IDS
WWW Server

IPS (Intrusion Prevention System)

IPS (Intrusion Prevention System) IDS ka advanced version hai jo **attack detect karne ke saath-saath usse automatically block bhi karta hai**.

☞ IPS **inline mode** me kaam karta hai (traffic ke beech me lagta hai).

Working



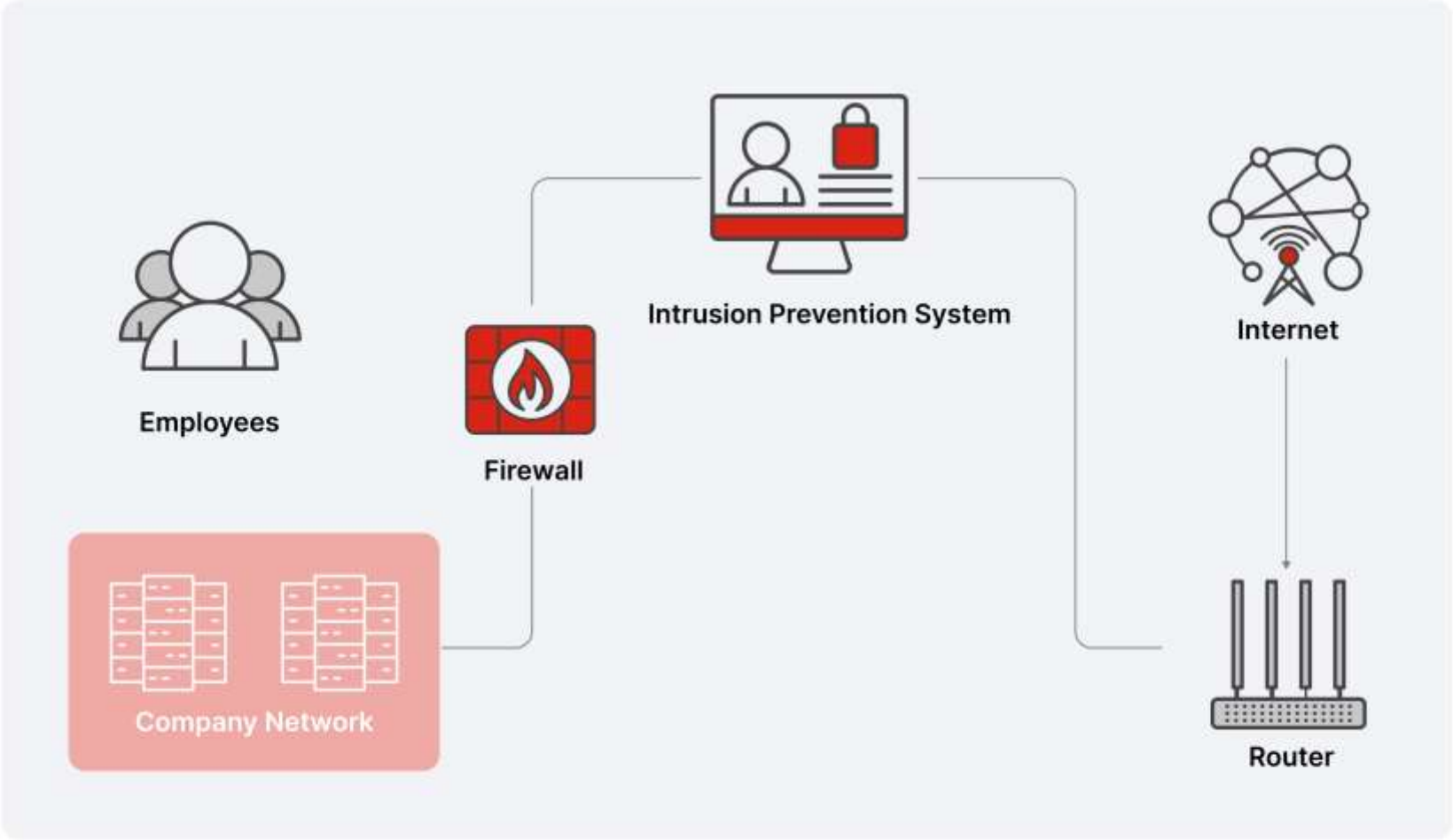
1.Traffic IPS ke through pass hota hai
2.Malicious activity detect hoti hai
3.IPS action leta hai:
    1. Packet drop
    2. IP block
    3. Connection reset

Example

Agar SQL Injection attack detect hota hai:

•IPS turant request ko **block kar dega**
•Attacker ka IP blacklist kar sakta hai

What Is an Intrusion Prevention System (IPS)?



🔍 Detection Techniques

IDS/IPS attacks ko detect karne ke liye different techniques use karte hain:

1. Signature-Based Detection

Is technique me system **known attack patterns (signatures)** ko database se match karta hai.

Example

- Known malware hash match
- Known SQL injection pattern



Advantages

- Accurate detection for known attacks
- Low false positives

Disadvantages

- New (zero-day) attacks detect nahi karta

2. Anomaly-Based Detection

Is technique me system **normal behavior ka baseline banata hai** aur usse compare karta hai.
👉 Agar behavior abnormal ho → alert

Example
•Normal user 9 AM login karta hai
•Suddenly 3 AM unusual login → suspicious



Advantages
•Unknown attacks detect kar sakta hai
•Zero-day detection possible
Disadvantages
•High false positives
•Training required

3. Stateful Protocol Analysis

Is technique me system **protocol behavior (HTTP, FTP, TCP)** ko analyze karta hai.
☞ Agar protocol rules violate ho → attack detect

Example
•Invalid TCP handshake
•Malformed HTTP request



Advantages
•Protocol-level detection strong hota hai
•Better than signature-only
Disadvantages
•Resource intensive
•Complex implementation